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Yuhei Hara, Shota Furuta, Ayako Sukegawa, Koichiro Fujimaki & Shinya Hisano

Title

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Names and affiliations

Yuhei Hara, OTR., MS., Lecturer

Department of Occupational Therapy, Chiba Medical Welfare College

1019 Kamiyue, Kimitsu City, Chiba 299-1138, Japan

Second Stage of Doctoral Program, Graduate School of Comprehensive Scientific Research, Prefectural University of Hiroshima

1-1 Gakuen-cho, Mihara City, Hiroshima 723-0053, Japan

otlifehack@gmail.com

Shota Furuta, OTR., Lecturer

Department of Occupational Therapy, Rehabilitation College Shimane

2086-1 Furuichiba, Misumi-cho, Hamada City, Shimane 699-3226, Japan

furuta@rcs.ac.jp

Ayako Sukegawa, OTR., Ph.D., Associate Professor

Graduate School of Comprehensive Scientific Research, Prefectural University of Hiroshima

1-1 Gakuen-cho, Mihara City, Hiroshima 723-0053, Japan

a-sukegawa@pu-hiroshima.ac.jp

Koichiro Fujimaki, MD., Ph.D., Professor

Graduate School of Comprehensive Scientific Research, Prefectural University of Hiroshima

1-1 Gakuen-cho, Mihara City, Hiroshima 723-0053, Japan

fujimaki@pu-hiroshima.ac.jp

Shinya Hisano, OTR., Ph.D., Professor

Graduate School of Comprehensive Scientific Research, Prefectural University of Hiroshima

1-1 Gakuen-cho, Mihara City, Hiroshima 723-0053, Japan

hisano@pu-hiroshima.ac.jp

Corresponding author

Shinya Hisano, OTR., Ph.D., Professor

Graduate School of Comprehensive Scientific Research, Prefectural University of Hiroshima

1-1 Gakuen-cho, Mihara City, Hiroshima 723-0053, Japan

E-mail: hisano@pu-hiroshima.ac.jp

Abstract

Burnout is increasingly recognized as affecting students as well as workers, prompting the development of numerous self-report measures to assess burnout across diverse student populations. Despite the growing recognition of student burnout as a significant mental health concern, the methodological quality of instruments designed to measure this construct in student populations has not been rigorously and systematically evaluated using contemporary psychometric standards. This systematic review sought to identify and evaluate existing self-report measures used to measure student burnout, appraising their measurement properties and current limitations. Adhering to PRISMA and COSMIN guidelines, we examined studies that developed or validated self-report measures of student burnout. Forty-four studies met the inclusion criteria, yielding 33 distinct instruments. Quality appraisal indicated that ten Self-report measures warranted a 'Conclude with recommendation against its use.' Strikingly, no instrument met the COSMIN criteria for a recommendation for use. These findings underscore the urgent need for de novo instrument development or substantial revision of existing measures, beginning with rigorous content validity studies as mandated by COSMIN guidelines, given the absence of currently recommended tools. Such work is essential to enable evidence-based interventions that effectively address the mental health impacts of burnout among diverse student populations.

Keywords:

Burnout, Self-report measures, Students, Cosmin

Introduction

Burnout, originally conceptualized by Freudenberger in the 1970s [1], is defined in the International Statistical Classification of Diseases and Related Health Problems, 11th Revision (ICD-11), published by the World Health Organization (WHO) [2], as "Burnout is a syndrome conceptualized as resulting from chronic workplace stress that has not been successfully managed."

Research on burnout has traditionally focused on employees, particularly those in the helping professions, suggesting that academic engagement is often considered equivalent to professional work for students, and that student burnout is thought to manifest through academic activities [3, 4]. Prior investigations into burnout prevalence among university students indicate that a substantial proportion exhibit burnout tendency [5], which have been linked to dropout intentions [6], academic performance outcomes [6, 7], suicidal ideation [8], and the prediction of depressive symptoms [9]. These findings underscore the urgent need for burnout prevention initiatives not only among workers but also among students. Moreover, healthcare students enrolled in healthcare-related disciplines (e. g., occupational therapy, nursing, physical therapy) commonly experience elevated stress levels and compromised mental health [10-14]. A longitudinal study involving healthcare students (occupational therapy, nursing, psychology, and social work) demonstrated that exhaustion and cynicism were highest during the final academic year, followed by the second year of employment, while professional efficacy was lowest in the final year and gradually improved thereafter [15].

These findings indicate that burnout is a significant concern in healthcare-related professions and warrants proactive attention beginning in the student phase. Consequently, research on coping mechanisms and intervention strategies has progressed, with studies investigating burnout-related correlates among healthcare students [14] and evaluating preventive interventions targeting burnout [16].

Current Issues in Measuring Student Burnout

To rigorously evaluate and mitigate this syndrome, the utilization of psychometrically robust measurement instruments is indispensable. Existing burnout scales for students comprise a combination of student-specific measures, such as the MBI-SS (Maslach Burnout Inventory-Student Survey), and worker-oriented instruments, such as the MBI (Maslach Burnout Inventory) [14]. The MBI [17] has been translated into multiple languages [18-20], and student-focused instruments, including the MBI-SS, have been developed and are widely implemented internationally. However, concerns regarding the psychometric properties of student burnout measures have been documented. Specifically, problems with the internal consistency of certain subscales—particularly depersonalization—have been consistently reported [21]. Furthermore, the factorial validity of instruments adapted from occupational contexts has been questioned. Maroco & Campos [22] reported that the variance of the MBI-SS factors was low, suggesting that the scale may not accurately reflect the burnout structure in student samples. Additionally, Tsubakita & Shimazaki [23] reported that the three-factor model of the Japanese version of the MBI-SS did not demonstrate sufficient structural validity, and that a bifactor model produced better results. Based on these findings, we concluded that the MBI-SS did not reproduce the expected three-factor structure in student samples [22, 23]. Notably, a recent systematic review of occupational burnout measures using COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) methodology concluded that no instrument could be recommended for use [24]. This finding from the occupational literature provides a compelling rationale for examining whether student-specific instruments fare any better. Although these findings underscore the necessity for further burnout research targeting students, validity and reliability issues persist for the MBI, the most employed burnout instrument globally. Accordingly, it is essential to critically examine the MBI-SS alongside other measures developed specifically for student populations.

Burnout is classified as a syndrome in ICD-11, and student respondents are not patients in the clinical sense. COSMIN guidelines were developed for evaluating Patient-Reported Outcome Measures (PROMs) in clinical populations. However, given that burnout is a recognized health-related syndrome, that measures of burnout are used to screen for health risk and evaluate interventions, and that a previous COSMIN-based review of occupational burnout instruments [24] established precedent, we judged the application of COSMIN methodology to be appropriate for this review. Nonetheless, readers should interpret the findings with this conceptual consideration in mind.

Rationale and Objectives of This Review

Given that even the most widely used instrument (the Maslach Burnout Inventory) has well-documented psychometric limitations in student populations, and that a previous COSMIN-based review found no recommended measures for occupational burnout [24], a systematic appraisal of student-specific instruments is not merely warranted but essential. Therefore, this systematic review aimed to: (1) identify all self-report measures developed or adapted to measure burnout in student populations; (2) critically appraise, synthesize, and compare the measurement properties of these instruments using the COSMIN methodology; (3) provide recommendations for the selection of student burnout measures for research and practice; and (4) classify instruments according to COSMIN recommendation categories (A, B, C).

Method

Protocol design and registration

This study was conducted as a systematic review, with the analysis of self-report measures performed in accordance with the COSMIN guideline for systematic reviews of patient-reported outcome measures version 2.0 [25]. Reporting adhered to the PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews [26]. This review was registered on PROSPERO (ID: CRD42025635617).

Source of data

The search strategy was developed using combinations of keywords related to burnout, student populations, psychometric properties, and measurement instruments. The complete search string was as follows: ("burnout" OR “burnout syndrome”) AND ("students" OR "student" OR "academic" OR "school” OR "learning"

OR “undergraduate”) AND ("validity" OR "reliability" OR “content” OR “Structural” OR “Internal” OR “Cross-cultural” OR “Measurement invariance” OR “Measurement error” OR “Hypotheses testing for construct validity” OR “responsiveness” OR “minimal detectable change” OR “minimal clinically important difference” OR “factor analysis” OR “validation” OR “psychometric” OR “interpretability” OR “feasibility” OR “Measurement”) AND (“questionnaire” OR “patient-reported outcome” OR “scale” OR “Inventory”).

Selection criteria

Original articles reporting the development and/or validation of self-report measures designed to assess burnout among students were considered for inclusion. Eligible studies had to meet the following criteria: (1) the study focused on the development or psychometric evaluation of a self-report measure for student burnout; (2) the original version of the self-report measure (not a translated version) was examined in at least one study; (3) the article was an original research paper; (4) the target population consisted of students (no restrictions on student characteristics); (5) the article was written in English; and (6) full-text access was available. The following exclusion criteria were applied: (1) full text was not accessible; (2) articles written in languages other than English; (3) non-original research such as conference proceedings, commentaries, or review papers; (4) studies that did not report student characteristics.

Review process

The searches were conducted from March 1 - April 17, 2025, in each search engine separately: MEDLINE(PubMed), EMBASE(Embase), Web of Science, ERIC, CINAHL(EBSCOhost), PsycINFO (Ovid), and Cochrane Library. The search period extended from the inception of each database to March 31, 2025, encompassing MEDLINE (1946–present), EMBASE (1974–present), Web of Science (1965–present), ERIC (1907–present), CINAHL (1981–present), PsycINFO (1800–present), and the Cochrane Library (1993–present). If a systematic review was identified through the literature search, studies meeting the eligibility criteria were retrieved from its reference list. Additionally, hand searches (Subsequently, the reference lists of the included articles were searched to identify further studies) were performed.

The full list of articles retrieved from the seven search engines and hand searches was saved in MS Excel for data management and synthesis, eliminating duplicates. Two reviewers (YH, SH) independently performed Data Extraction and assessed the full texts of all studies that met the eligibility criteria based on title and abstract screening. Before beginning the screening process, the reviewers held meetings, conducted calibration exercises, and discuss and clarify the eligibility and exclusion criteria. Furthermore, reviews carefully read and fully understood the COSMIN guidelines before proceeding. Disagreements were resolved through discussion or consultation with a third reviewer (SF).

The number of included and excluded articles was documented at each stage using the PRISMA flow diagram [26] to depict the study selection process, specifying the records identified, screened, excluded, and ultimately incorporated into the final review.

The characteristics of studies featured in prior processes were synthesized, including author and publication year, participant characteristics, sample size, gender distribution, mean age, institutional affiliation, the self-report measure employed, its language, subscales, number of items, mode of subjective assessment, psychometric properties evaluated, and test–retest interval.

Subsequently, the methodology used to assess each study's measurement properties was appraised using a risk-of-bias checklist [25]. The results pertaining to each measurement property were then evaluated according to established criteria for good measurement properties [25]. Finally, the methodological quality and findings of each study were integrated for each self-report measure, and the overall evidence for each self-report measure was graded using a modified GRADE approach [25] to determine which instruments should be recommended for use. Although the ICD-11 employs subscales of the MBI as the conceptual definition of burnout, there is no empirical justification for treating the MBI as a gold standard; therefore, criterion-related validity was not evaluated in this study.

The methodological quality of the measurement properties in each included study was evaluated using a risk-of-bias checklist employing a four-point rating scale: very good, adequate, doubtful, or inadequate. A score was assigned for each box (from highest to lowest: very good, adequate, doubtful, inadequate, N/A), and a summary table was generated. The lowest rating among the sub-items within each box was adopted as the final score for that box. Subsequently, the results reported in each study were appraised using the updated criteria for good measurement properties, with each outcome rated on a three-point scale: “+” (sufficient), “–” (insufficient), or “?” (indeterminate) [25].

After completing the evaluation of each study, a modified GRADE approach was applied based on the measurement properties of each study, as measured using the criteria for good measurement properties [25]. The overall measurement properties of the self-report measure were rated on a four-point scale: "sufficient (+)", "insufficient (–)", "inconsistent (±)", or "indeterminate (?)". The quality of the self-report measure's evidence was also assessed on a four-point scale: "high", "moderate", "low" or "very low".

The measurement property ratings and quality of evidence for each self-report measure were summarized in a Summary of Findings (SoF) Table. The results of each study were used to rate the self-report measure's evidence, and the quality of the evidence was then assessed using the GRADE approach. It has been suggested that imprecision should not be assessed for content validity [25], and this study follows this approach. The GRADE approach was used to assess risk of bias, inconsistency, imprecision, and indirectness in accordance with the COSMIN guidelines version 2. 0 [25]. In accordance with the COSMIN guidelines version 2.0, formulate conclusions for recommendations per self-report measure was used based on the following criteria: 1) Conclude with recommendation for use: High-quality evidence confirmed for all measurement properties; 2) Conclude with recommendation against its use: Insufficient high-quality evidence confirmed for one or more measurement properties; 3) No conclusion: Any rating other than 1) or

2). In addition, if there was no recommended self-report measure, the system was classified into three categories using the criteria of the COSMIN guidelines [27]. A (self-report measures recommended for use): There is sufficient content validity and at least low-quality evidence for sufficient internal consistency. B (self-report measures not classified as A or C): Further validation is required. C (self-report measures that should not be recommended for use): There is high-quality evidence that they have insufficient measurement properties.

Results

Selected studies

From the initial search, 13,719 records were identified. Following title and abstract screening, 146 articles were retained for full-text review. In this primary screening, there were 14 disagreements, and therefore the resolution rate was also 99%. After, there were no disagreements. Ultimately, 40 studies met the eligibility criteria and were included in this systematic review. Additionally, four references related to self-report measure development were identified within the included articles and incorporated through manual searching by reviewing reference lists of included articles, resulting in a total of 44 studies included in the review. The study selection process for the systematic review is presented in Figure 1.

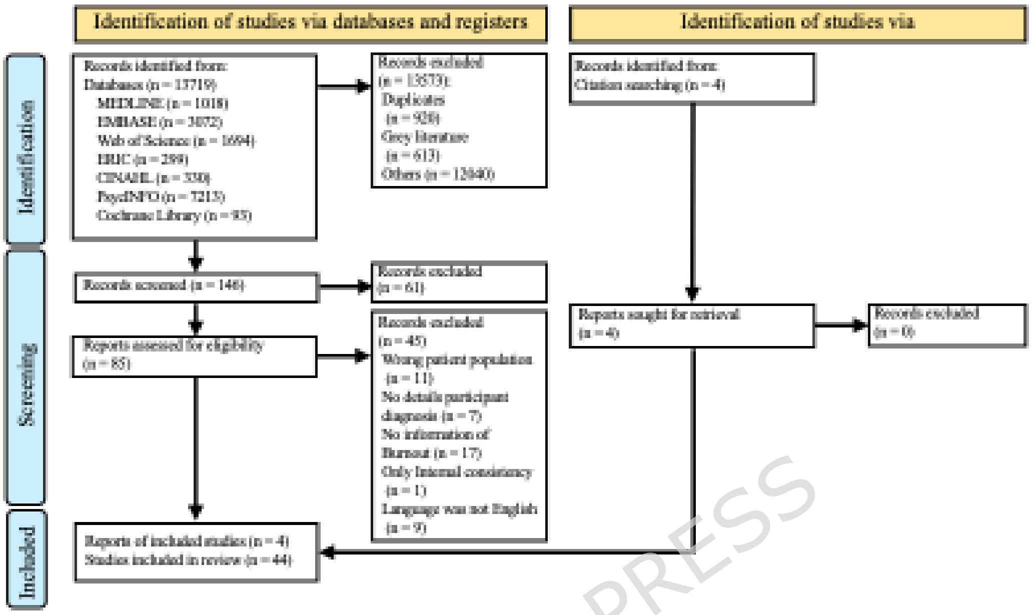

Figure 1 PRISMA 2020 flow diagram

The studies included in this systematic review assessed 44 studies. This review of 44 studies examined measurement properties of 33 self-report measures. The most frequently examined properties were structural validity (43 evaluations) and internal consistency (38 evaluations), whereas no studies investigated measurement error or responsiveness. Comprehensive information on study characteristics, the populations employed for the assessment of measurement properties, and the intervals between retests is provided in Additional file 1.

The self-report measures identified comprised the following instruments: Academic

Burnout Scale [28]; ASBI (Adolescent Student Burnout Inventory) [29, 30]; BAT-C Student Version (Burnout Assessment Tool) [31]; BCSQ-12-SS (12-Item Burnout Clinical Subtype Questionnaire) [32]; CBI-S (Copenhagen Burnout Inventory – Student Version) [22, 33, 34]; E-learning Burnout Scale [35]; ESSBS (Elementary Students School Burnout Scale for Grades 6–8) [36]; FS-BAT (Flemish School Burnout Assessment Tool) [37]; MBI [38]; MBI (student-adapted version) [39]; MBI-EFL-SS (Maslach Burnout Inventory – EFL Student Survey) [40]; MBI-ES (Maslach Burnout Inventory – Educators Survey) [41]; MBI-ES revised [42]; MBI-GS (Maslach Burnout Inventory – General Survey, student-adapted) [43]; MBI-HSS (Maslach Burnout Inventory – Human Services Survey, modified) [44]; MBI-SS (Spanish version) [3, 45-48]; MBI-SS (Netherlands version) [3]; MBI-SSi (Maslach Burnout Inventory – Student Survey with reversed efficacy dimension) [49, 50]; MBI-SS modified target terms [51]; MBI-SS with negatively worded items and one item removed (Dutch); MBI-SS with negatively worded items and one item removed (Spanish) [52]; MBI-SuS (Maslach Burnout Inventar – Version für Schülerinnen und Schüler) [53]; OLBI-S (Oldenburg Burnout Inventory – Student Version) [22, 33, 54]; OLBI-N (Oldenburg Burnout Inventory in Nursing) [55]; S-BAT (Student Version of the Burnout Assessment Tool) [56]; SBI (School Burnout Inventory) [57-59]; SBI-4

[60]; SBI-8 [61-63]; SBI-U [64]; Short Version of the Emotional Burnout Assessment Instrument (BAT-C) [65]; SSBS (Secondary School Burnout Scale) [66]; ULB (Undergraduate Learning Burnout) Scale [67,68]; and UW-ESBS revised (University of Windsor Employed Student Burnout Survey, revised) [69] (Additional file 1). Regarding the MBI-SS, both the Spanish and Dutch versions were developed within the same study, and this review adopted both as the original versions. Furthermore, several self-report measures, the modified version of the MBI-SS, were developed to address existing issues with the MBI-SS. Since these scales do not fully align with the MBI-SS in terms of content, this review evaluated them as separate instruments. As with the MBI-SS, several revised self-report versions of the MBI, SBI, and OLBI have been developed, and this review evaluated them as separate measures.

Target populations (with overlap) included: elementary school students (2 instruments, 6%), middle school students (4.12%), high school students (7.21%), and university students (27.82%), of whom six studies (18%) involved healthcare students.

Table 1 Summary of ratings using COSMIN Risk-of-bias checklist and criteria for good measurement properties

	COSMIN Risk-of-bias checklist					Criteria for good measurement properties				
	Very good	Adequate	Doubtful	Inadequate	Not evaluated	sufficient	Insufficient	indeterminate	inconsistent	Not evaluated
PROM development	0(0%)	0(0%)	23(49%)	2(4%)	22(43%)	0(0%)	0(0%)	25(53%)	0(0%)	22(47%)
Content validity	0(0%)	0(0%)	4(9%)	0(0%)	43(91%)					
Structural validity	32(68%)	1(2%)	7(15%)	5(11%)	2(4%)	19(40%)	17(36%)	9(19%)	0(0%)	2(4%)
Internal consistency	34(72%)	0(0%)	6(13%)	0(0%)	7(15%)	11(23%)	5(11%)	24(51%)	0(0%)	7(15%)
Cross-cultural validity/Measurement invariance	0(0%)	0(0%)	16(34%)	1(2%)	30(64%)	2(4%)	7(15%)	8(17%)	0(0%)	30(64%)
Reliability	0(0%)	0(0%)	2(4%)	2(4%)	43(91%)	1(2%)	2(4%)	1(2%)	0(0%)	43(91%)
Measurement error	0(0%)	0(0%)	0(0%)	0(0%)	47(100%)	0(0%)	0(0%)	0(0%)	0(0%)	47(100%)
Hypotheses testing for construct validity	7(15%)	4(9%)	4(9%)	16(34%)	16(34%)	1(2%)	5(11%)	14(30%)	11(23%)	16(34%)
Responsiveness	0(0%)	0(0%)	0(0%)	0(0%)	47(100%)	0(0%)	0(0%)	0(0%)	0(0%)	47(100%)

Note. According to the COSMIN guideline, "PROM development" and "content validity" are assessed together under the criteria for good content validity.

As shown in Table 1, methodological and quality, as assessed by the COSMIN Risk-of-bias checklist, varied widely across the 44 included studies. A common issue across many studies was "doubtful" or "inadequate" ratings for "PROM development" (row 1) and "Content validity" (row 2), as assessed by indicating foundational flaws in the initial construction of these instruments. In contrast, "Structural validity" (row 3) and "Internal consistency" (row 4) were more frequently rated as "adequate" or "very good," reflecting the field's historical emphasis on these psychometric properties. However, "Responsiveness" (row 7) and "Measurement error" (row 9) were unevaluated, representing critical gaps for longitudinal research applications.

Results of the psychometric properties of self-report measures

Comprehensive details concerning the appraisal of synthesized findings and the evaluation of evidence quality for each self-report measure are presented in

Additional file 2.

Content validity was examined across 24 self-report measures; however, the available information was insufficient to comprehensively evaluate measurement properties for all instruments. A substantial proportion of the self-report measures identified in this study were adapted for assessing student burnout by modifying the wording of instruments originally developed for occupational burnout, replacing references to “work” with “school” or “study”.

Structural validity was assessed for 32 self-report measures. Evaluation of structural validity revealed a clear pattern distinguishing instrument types. High-quality evidence for sufficient structural validity was predominantly found in newer instruments developed specifically for student populations, particularly adaptations of the Burnout Assessment Tool (BAT-C Student Version, S-BAT, Short version of BAT-C). In contrast, adaptations of the Maslach Burnout Inventory frequently demonstrated structural problems, with confirmatory factor analyses failing to replicate the expected three-factor structure across multiple studies and contexts. This pattern suggests that instruments grounded in contemporary burnout conceptualizations (the four-dimensional BAT model) may possess superior factorial validity compared to adaptations of the three-dimensional MBI framework.

Internal consistency was evaluated for 29 self-report measures. Nine instruments demonstrated satisfactory measurement properties with high-level evidence: Academic Burnout Scale; BAT-C Student Version; FS-BAT; MBI-EFL-SS; MBI-SS (Spanish version); MBI-SuS; OLBI-N; S-BAT; and the Short Version of the Emotional Burnout Assessment Instrument. In contrast, three self-report measures—MBI-SS (Dutch version), SBI, and SSBS—were identified as having inadequate measurement properties despite high-level evidence.

Cross-cultural validity and measurement invariance were examined for 12 self-report measures. None demonstrated both adequate measurement properties and high-level evidence, nor were any found to have inadequate properties with high-level evidence. Two instruments, S-BAT and SBI-8, were rated "+" for measurement properties but supported only by low-level evidence. Notably, MBI-SS (Spanish version) achieved a "+" rating for measurement invariance but a "–" rating for cross-cultural validity, with both supported by low-level evidence (Table 2).

Table 2 Summary of Findings Table about Evaluation of self-report measures and formulation of recommendations for use

Self-report measure / Measurement property	Content validity		Structural validity		Internal consistency		CCV / MI		Reliability		HTCV		Recommendations	Categories
	Evid.	Rate	Evid.	Rate	Evid.	Rate	Evid.	Rate	Evid.	Rate	Evid.	Rate		
Academic burnout scale		?	H	+	H	+						?	NC	B
ASBI				±		?	M	−					NC	B
BAT-C-student version		?	H	+	H	+						±	NC	B
BCSQ-12-SS		?	M	−		?					H	−	NR	C
CBI-S		?	M	−		?		?			M	−	NC	B
e-learning burnout				?		?						?	NC	B
ESSBS		?	H	−		?					VL	−	NR	C
FS-BAT		?	H	+	H	+			L	−	H	+	NC	B
MBI				?		?		?					NC	B
MBI (student-adapted version)		?		?		?						±	NC	B
MBI-EFL-SS		?	H	+	H	+			VL	−		?	NC	B
MBI-ES				?		?							NC	B
MBI-ES revised		?							VL	+			NC	B
MBI-GS revised		?	VL	−		?						±	NC	B
MBI-HSS revised				?		?							NC	B
MBI-SS (ver Spanish)		?	H	+	H	+	L	− / +				±	NC	B
MBI-SS (ver Dutch)		?	H	+	H	−	L	−				±	NR	C
MBI-SSi		?	H	−		?						?	NR	C
MBI-SS (modified target terms)		?		?									NC	B
MBI-SS (Negative removed: ver Dutch)		?	H	−		?	L	−					NR	C
MBI-SS (Negative removed: ver Spanish)		?	H	−		?	L	−					NR	C
MBI-SuS			H	+	H	+	L	−				±	NC	B
OLBI-S		?		±	M	−		?			M	−	NC	B
OLBI-N		?	H	+	H	+					H	−	NR	C
S-BAT		?	H	+	H	+	L	+				±	NC	B
SBI			L	+	H	−		?				±	NR	C
SBI-4				?							L	+	NC	B
SBI-8			H	+		±	L	+	VL	−		?	NC	B
SBI-U		?	H	+	L	+					H	−	NR	C
Short version of BAT-C		?	H	+	H	+						?	NC	B
SSBS		?	H	+	H	−						±	NR	C
ULB scale		?	H	+	L	−							NC	B
UW-ESBS revised		?	H	+								?	NC	B

Note. CCV / MI = Cross−cultural / Measurement invariance, HTCV = Hypotheses testing for construct validity, Measurement invariance, Evid. = Evidence, Rate = Ratiing, H = High, M = Moderate, L = Low, VL = Very Low, "+" = sufficient, "−" = insufficient, "?" = indeterminate, "±" = inconsistent, NC = No conclusion, NR = Not recommended, B = Self-report measures have potential to be recommended for use, but they require further research to assess the quality of these Self-report measures (There is No evidence for sufficient content validity (any level), and categorized not in C), C = Self-report measures should not be recommended for use (There is high-quality evidence that they have insufficient measurement properties), Negative removed = negatively worded items and one item removed, ASBI = Adolescent Student Burnout Inventory, BAT = Burnout Asessment tool, BCSQ-12-SS = The 12-question burnout clinical subtype questionnaire, CBI-S = Copenhagen Burnout Inventory - Student Version, e-learning burnout, ESSBS = Elementary Students School Burnout Scale for Grades 6-8, FS-BAT = A validation of the Flemish School Burnout Assessment Tool for students, MBI = Maslach Burnout Inventory, MBI-EFL-SS = Maslach Burnout Inventory-EFL Student Survey, MBI-ES = Maslach Burnout Inventory - Educators Survey, MBI-GS = Maslach Burnout Inventory-General Survey, MBI-HSS = Maslach Burnout Inventory - Human Services Survey, MBI-SS = Maslach Burnout Inventory-Student Survey, MBI-SSi = Maslach Burnout Inventory - Student Survey with the efficacy dimension reversed, MBI-SuS = Maslach Burnout Inventar–Version für Schülerinnen und Schüler, OLBI-S = Oldenburg Burnout Inventory – Student Version, OLBI-N = Oldenburg Burnout Inventory in Nursing, S-BAT = Student version of the Burnout Assessment Tool, SBI = School-Burnout Inventory, SBI-U = School Burnout Inventory-Universitarios, Short version of BAT-C = Short version of the emotional burnout assessment instrument (BAT-C), SSBS = Secondary School Burnout Scale, ULB = Undergraduate Learning Burnout scale, UW-ESBS = University of Windsor Employed Student Burnout Survey

Reliability was assessed for four self-report measures. No instruments exhibited both adequate measurement properties and high-level evidence, nor inadequate properties with high-level evidence. Only one self-report measure, MBI-EFL-SS, was rated “+” for measurement properties, supported by very low-level evidence (Table 2).

Hypothesis testing for construct validity was conducted for 24 self-report measures. FS-BAT was the sole instrument demonstrating adequate measurement properties with high-level evidence. Conversely, three self-report measures—BCSQ-12-SS, OLBI-N, and SBI-U—were identified as having inadequate measurement properties despite high-level evidence (Table 2).

Discussion

This study systematically appraised the measurement properties of self-report measures designed to assess burnout among students. To ensure methodological rigor, the review adhered to the COSMIN guidelines version 2. 0 [25] for systematic evaluations of self-report measures.

The review encompassed 44 studies and identified 33 distinct self-report measures. Of these, ten instruments were classified as “Conclude with recommendation against its use” (Category C), while 23 were categorized as “No Conclusion”

(Category B), indicating the absence of any high-quality self-report measure currently available for assessing student burnout. The principal finding of this systematic review—that no student burnout measure meets COSMIN criteria for recommendation for use—aligns with and extends the conclusions of Shoman et al. [24], who reported a similar absence of recommendable instruments for occupational burnout. This convergence suggests that the psychometric limitations observed in occupational burnout measurement may extend to student populations, possibly due to the reliance on MBI-derived instruments in both domains: The findings suggest a pattern in which the proliferation of instruments has outpaced rigorous psychometric evaluation, particularly regarding foundational properties such as content validity.

Four instruments—the BAT-C Student, S-BAT, Short version of BAT-C, and Academic burnout scale—demonstrated partially good psychometric properties across multiple domains, including structural validity, internal consistency, and hypothesis testing for construct validity. However, a critical caveat must be emphasized: content validity remained inadequately addressed for all four instruments. Under COSMIN guidelines, content validity is prioritized as the most essential measurement property, as an instrument cannot validly measure a construct if its items do not adequately represent that construct's domain. Consequently, these instruments cannot be recommended for use in their current form. Rather, they should be considered potentially suitable for further development, with urgent priority given to comprehensive content validity studies involving student populations to ensure that item content authentically captures the student burnout experience.

Among the ten self-report measures classified as “Not Recommended” (Category C), four were adaptations of the MBI with wording modified for student populations. A meta-analysis of studies employing the MBI revealed that numerous depersonalization items failed to achieve a Cronbach’s α coefficient exceeding 0. 70 [21]. And Beer & Bianchi [70] suggest that personal accomplishment might be understood as a distinct construct rather than a core component of burnout, suggesting potential areas for refinement in the original MBI framework.

Notably, none of the self-report measures identified as potentially recommendable in this review were derived from the MBI, underscoring the need to reconsider the suitability of MBI-based instruments for assessing student burnout. Conversely, three of the none of the self-report measures deemed potentially recommendable were adapted from the BAT, originally developed by Schaufeli et al. [3] for occupational burnout, with wording tailored to the student context. Unlike the MBI, which was developed inductively from human services samples and may not translate well to student contexts, the BAT was developed with explicit attention to conceptual clarity and has demonstrated superior structural validity across diverse populations.

The superior psychometric performance of BAT-derived instruments compared to MBI adaptations is theoretically significant. As Schaufeli et al. [71] articulated, the BAT was developed based on a contemporary reconceptualization of burnout as a multidimensional syndrome comprising exhaustion, mental distance, cognitive impairment, and emotional impairment. They further delineated these four core dimensions as “accompanied by depressive mood and nonspecific psychological and psychosomatic complaints,” explicitly clarifying that the term ‘work’ is not confined to paid employment but also encompasses academic contexts. In contrast, the MBI's three-factor structure (exhaustion, cynicism, reduced efficacy) was developed inductively in the 1980s based on human services workers [17]. The persistent structural problems observed in MBI adaptations may therefore reflect a fundamental conceptual misfit rather than merely methodological limitations. This suggests that future instrument development should prioritize theoretically grounded, student-centered frameworks over adaptations of occupational measures. In addition, we should reconsider continued reliance on MBI adaptations, and BAT framework supersedes MBI framework for development of new student burnout measures, as well as the re-evaluation of existing ones.

An important consideration is the heterogeneity of student populations included in this review, spanning elementary school children through postgraduate university students, and encompassing both general student samples and specific subgroups such as medical and healthcare students. The lack of recommendable instruments held true across all these populations, suggesting that the psychometric deficiencies are not confined to particular educational levels or disciplinary contexts. However, this heterogeneity also complicates interpretation, as the experience of burnout likely differs qualitatively between a 10-year-old elementary student and a 25-year-old medical resident. This heterogeneity likely precluded recommending a single instrument, as one tool is unlikely to function equivalently across diverse populations. Therefore, future research should examine developmental differentiation, stratifying burnout assessments by specific academic or training stages.

Therefore, this study emphasizes that the development of new student burnout measures, as well as the re-evaluation of existing ones, must begin with content validity research—the most critical property according to the COSMIN guidelines. Specifically, content validity studies should involve in-depth qualitative interviews with diverse student populations to ensure item content captures the lived experience of burnout across different educational levels and cultural contexts. Namely, this requires conducting rigorous qualitative research to ensure that the scale items accurately reflect the lived experiences of diverse target student populations.

Future Direction

Advancing research on student burnout necessitates the utilization of student-specific self-report measures that exhibit robust measurement properties and high-quality content validity. The findings of this review indicate that no existing self-report measure for student burnout can currently be strongly recommended for use. Consequently, the absence of a gold standard that can be strongly recommended for use remains a major issue in the field of student burnout research. This underscores the need for rigorous re-evaluation of existing instruments using high-quality methodological approaches, as well as the development of novel self-report measures tailored to the student population. Moreover, the heterogeneity identified in this review poses issues for instrument application. Future studies should begin with comprehensive content validity assessments, followed by evaluations of measurement invariance across educational levels and cultural contexts, other psychometric properties in accordance.

Strengths and Limitations

The COSMIN guidelines were originally developed as a methodological framework and set of practical tools for evaluating the quality of patient-reported outcome measures (PROMs) with the aim of identifying the most appropriate instruments. We defined the measure used in this study assesses health-related burnout, the study was conducted using the COSMIN guidelines based on Shoman et al. [24]. However, because burnout is classified as a syndrome in the ICD-11 [2], the student participants in our study may not be considered patients. Consequently, applying the COSMIN criteria in this context may risk excluding potentially valuable measurement instruments, as the criteria may be overly stringent for non-patient populations. Another limitation regarding the use of the COSMIN guidelines is that this review was conducted by researchers who had not received specialized training on COSMIN application. This may have influenced the methodology and the interpretation of the review results.

Other limitations of this review include the potential for publication bias, as studies reporting inadequate measurement properties of self-report measures are less likely to be published, and a potential source of selection bias due to the exclusion of otherwise valid psychometric studies that omitted demographic details. Additionally, restricting the analysis to English-language publications may have excluded instruments developed in other linguistic contexts. That is, these reviews may not have fully covered grey literature or instruments from non-English-speaking countries, and these may have been underrepresented. Moreover, this review included instruments measuring 'school burnout', 'academic burnout' and 'learning burnout' as conceptually equivalent to 'burnout' without evaluating whether they represent interchangeable terms or distinct constructs.

In addition, Cross-cultural validity and measurement invariance were insufficiently evaluated across the included studies, limiting the interpretability and comparability of burnout scores across cultural and educational contexts. Future research should prioritize rigorous examination of measurement invariance to ensure that instruments function equivalently across diverse student populations. Furthermore, a notable imbalance in the evaluation of measurement properties was observed, with extensive evidence on structural validity and internal consistency contrasted by a near-complete absence of data on responsiveness and measurement error. This gap critically limits the suitability of existing instruments for longitudinal research and intervention studies, where the ability to detect change over time and distinguish true change from measurement error is essential. In the COSMIN guideline, Measurement error refers to the systematic and random error of an individual patient’s score that is not attributed to true changes in the construct to be measured when taking multiple measurements. Without an assessment of measurement error in the self-report measure, it is impossible to determine whether the differences in results observed in intervention studies fall within "the range of measurement error". In the COSMIN guideline, Responsiveness refers to the ability of a PROM to detect change over time in the construct to be measured. Without an assessment of responsiveness in the self-report measure, even if students are actually reducing their burnout through the program, the self-report measure may fail to detect this change, leading to an erroneous conclusion that the program is “ineffective.” Furthermore, one limitation of the current evidence base is that studies examining key validity measures—including content validity—have pooled across levels from different educational levels. Pooling data across levels may mask important differences. Therefore, it is recommended developmental differentiation in future research.

Nonetheless, the strengths of this systematic review lie in its preregistered protocol, comprehensive search across multiple large-scale databases, and strict adherence to COSMIN guidelines for evaluating self-report measure quality.

Conclusion

This systematic review reveals a fundamental and concerning gap in student burnout research: despite 33 instruments and 44 validation studies, no measure meets contemporary psychometric standards for recommendation for use. The critical deficiency lies in content validity—the most fundamental measurement property—which remains inadequately addressed across nearly all instruments. There is a pressing need for de novo instrument development or substantial revision of existing measures, beginning with rigorous content validity studies as mandated by COSMIN guidelines. Without psychometrically sound instruments that authentically capture the student burnout experience, efforts to understand, prevent, and intervene in this critical mental health concern will remain fundamentally limited.

Declarations

Abbreviations

BAT: Burnout Assessment Tool

BCSQ-12-SS: 12-Item Burnout Clinical Subtype Questionnaire

COSMIN: Consensus-based Standards for the selection of health Measurement INstruments

ICD-11: International Statistical Classification of Diseases and Related Health Problems, 11th Revision

MBI: Maslach Burnout Inventory

MS: Microsoft

OLBI: Oldenburg Burnout Inventory

PROMs: patient-reported outcome measures

SBI: School Burnout Inventory

SSBS: Secondary School Burnout Scale

WHO: World Health Organization

Ethics approval and consent to participate

This study is a literature review and has not received ethical approval.

Consent for publication

The results/data/figures in this manuscript have not been published elsewhere, nor are they under consideration (from you or one of your Contributing Authors) by another publisher, and all of the material is owned by the authors and/or no permissions are required.

Availability of data and materials

The datasets generated and/or analyzed during the current study are available in the supplementary material (Additional file 1, 2, 3) submitted with this manuscript.

Competing Interests

I declare that the authors have no competing interests as defined by BMC, or other interests that might be perceived to influence the results and/or discussion reported in this paper.

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Authors' contributions

Conceptualization, Shinya Hisano; Methodology, Shinya Hisano; Formal analysis, Yuhei Hara, Shinya Hisano, Shota Furuta; Investigation, Yuhei Hara, Shinya Hisano;

Writing-original draft preparation, Yuhei Hara; Writing-review and editing, Ayako Sukegawa, Koichiro Fujimaki, Shinya Hisano; Supervision, Shinya Hisano

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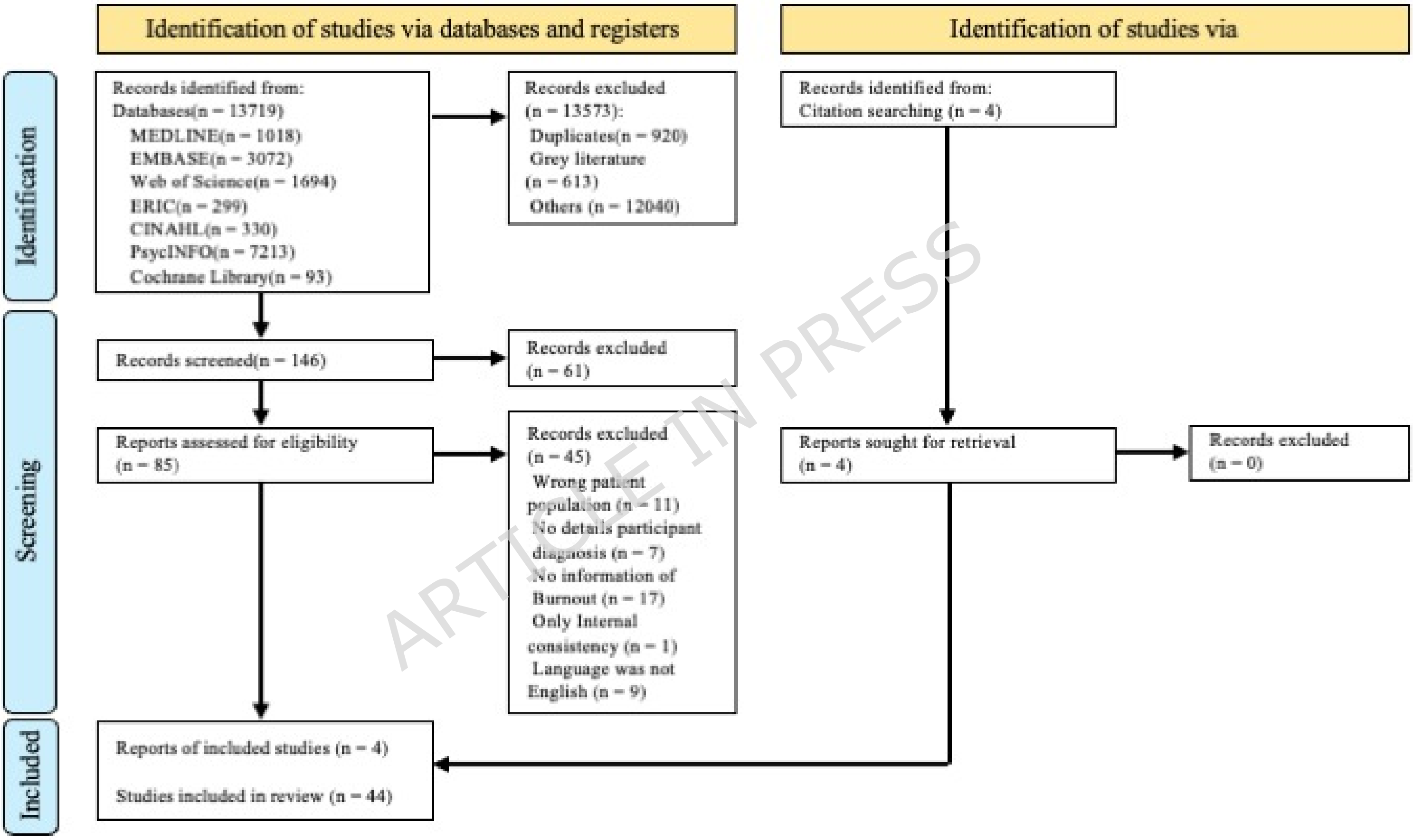

Identification of studies via databases and registers
Identification of studies via
Identification
Screening
Included
Records identified from:
Databases(n = 13719)
MEDLINE(n = 1018)
EMBASE(n = 3072)
Web of Science(n = 1694)
ERIC(n = 299)
CINAHL(n = 330)
PsycINFO(n = 7213)
Cochrane Library(n = 93)
Records excluded
(n = 13573):
Duplicates(n = 920)
Grey literature
(n = 613)
Others (n = 12040)
Records identified from:
Citation searching (n = 4)
Records screened(n = 146)
Records excluded
(n = 61)
Reports assessed for eligibility
(n = 85)
Records excluded
(n = 45)
Wrong patient
population (n = 11)
No details participant
diagnosis (n = 7)
No information of
Burnout (n = 17)
Only Internal
consistency (n = 1)
Language was not
English (n = 9)
Reports sought for retrieval
(n = 4)
Records excluded
(n = 0)
Reports of included studies (n = 4)
Studies included in review (n = 44)

Table 1 Summary of ratings using COSMIN Risk-of-bias checklist and criteria for good measurement properties

	COSMIN Risk-of-bias checklist					Criteria for good measurement properties				
	Very good	Adequate	Doubtful	Inadequate	Not evaluated	sufficient	insufficient	indeterminate	inconsistent	Not evaluated
PROM development	0(0%)	0(0%)	23(49%)	2(4%)	22(43%)	0(0%)	0(0%)	25(53%)	0(0%)	22(47%)
Content validity	0(0%)	0(0%)	4(9%)	0(0%)	43(91%)					
Structural validity	32(68%)	1(2%)	7(15%)	5(11%)	2(4%)	19(40%)	17(36%)	9(19%)	0(0%)	2(4%)
Internal consistency	34(72%)	0(0%)	6(13%)	0(0%)	7(15%)	11(23%)	5(11%)	24(51%)	0(0%)	7(15%)
Cross-cultural validity/Measurement invariance	0(0%)	0(0%)	16(34%)	1(2%)	30(64%)	2(4%)	7(15%)	8(17%)	0(0%)	30(64%)
Reliability	0(0%)	0(0%)	2(4%)	2(4%)	43(91%)	1(2%)	2(4%)	1(2%)	0(0%)	43(91%)
Measurement error	0(0%)	0(0%)	0(0%)	0(0%)	47(100%)	0(0%)	0(0%)	0(0%)	0(0%)	47(100%)
Hypotheses testing for construct validity	7(15%)	4(9%)	4(9%)	16(34%)	16(34%)	1(2%)	5(11%)	14(30%)	11(23%)	16(34%)
Responsiveness	0(0%)	0(0%)	0(0%)	0(0%)	47(100%)	0(0%)	0(0%)	0(0%)	0(0%)	47(100%)

Note. According to the COSMIN guideline, "PROM development" and "content validity" are assessed together under the criteria for good content validity.

Table 2 Summary of Findings Table about Evaluation of self-report measures and formulation of recommendations for use

Self-report measure / Measurement property	Content validity		Structural validity		Internal consistency		CCV / MI		Reliability		HTCV		Recommendations	Categories
	Evid.	Rate	Evid.	Rate	Evid.	Rate	Evid.	Rate	Evid.	Rate	Evid.	Rate		
Academic burnout scale		?	H	+	H	+						?	NC	B
ASBI				±		?	M	–					NC	B
BAT-C-student version		?	H	+	H	+						±	NC	B
BCSQ-12-SS		?	M	–		?					H	–	NR	C
CBI-S		?	M	–		?		?			M	–	NC	B
e-learning burnout				?		?						?	NC	B
ESSBS		?	H	–		?					VL	–	NR	C
FS-BAT		?	H	+	H	+			L	–	H	+	NC	B
MBI				?		?		?					NC	B
MBI (student-adapted version)		?		?		?						±	NC	B
MBI-EFL-SS		?	H	+	H	+			VL	–		?	NC	B
MBI-ES				?		?							NC	B
MBI-ES revised		?							VL	+			NC	B
MBI-GS revised		?	VL	–		?						±	NC	B
MBI-HSS revised				?		?							NC	B
MBI-SS (ver Spanish)		?	H	+	H	+	L	– / +				±	NC	B
MBI-SS (ver Dutch)		?	H	+	H	–	L	–				±	NR	C
MBI-SSi		?	H	–		?						?	NR	C
MBI-SS (modified target terms)		?		?									NC	B
MBI-SS (Negative removed: ver Dutch)		?	H	–		?	L	–					NR	C
MBI-SS (Negative removed: ver Spanish)		?	H	–		?	L	–					NR	C
MBI-SuS			H	+	H	+	L	–				±	NC	B
OLBI-S		?		±	M	–		?			M	–	NC	B
OLBI-N		?	H	+	H	+					H	–	NR	C
S-BAT		?	H	+	H	+	L	+				±	NC	B
SBI			L	+	H	–		?				±	NR	C
SBI-4				?							L	+	NC	B
SBI-8			H	+		±	L	+	VL	–		?	NC	B
SBI-U		?	H	+	L	+					H	–	NR	C
Short version of BAT-C		?	H	+	H	+						?	NC	B
SSBS		?	H	+	H	–						±	NR	C
ULB scale		?	H	+	L	–							NC	B
UW-ESBS revised		?	H	+								?	NC	B

Note. CCV / MI = Cross–cultural / Measurement invariance, HTCV = Hypotheses testing for construct validity, Measurement invariance, Evid. = Evidence, Rate = Ratiing, H = High, M = Moderate, L = Low, VL = Very Low, "+" = sufficient, "–" = insufficient, "?" = indeterminate, " ± " = inconsistent, NC = No conclusion, NR = Not recommended, B = Self-report measures have potential to be recommended for use, but they require further research to assess the quality of these Self-report measures (There is No evidence for sufficient content validity (any level), and categorized not in C), C = Self-report measures should not be recommended for use (There is high-quality evidence that they have insufficient measurement properties), Negative removed = negatively worded items and one item removed, ASBI = Adolescent Student Burnout Inventory, BAT = Burnout Asessment tool, BCSQ-12-SS = The 12-question burnout clinical subtype questionnaire, CBI-S = Copenhagen Burnout Inventory - Student Version, e-learning burnout, ESSBS = Elementary Students School Burnout Scale for Grades 6-8, FS-BAT = A validation of the Flemish School Burnout Assessment Tool for students, MBI = Maslach Burnout Inventory, MBI-EFL-SS = Maslach Burnout Inventory-EFL Student Survey, MBI-ES = Maslach Burnout Inventory - Educators Survey, MBI-GS = Maslach Burnout Inventory-General Survey, MBI-HSS = Maslach Burnout Inventory - Human Services Survey, MBI-SS = Maslach Burnout Inventory-Student Survey, MBI-SSi = Maslach Burnout Inventory - Student Survey with the efficacy dimension reversed, MBI-SuS = Maslach Burnout Inventar–Version für Schülerinnen und Schüler, OLBI-S = Oldenburg Burnout Inventory – Student Version, OLBI-N = Oldenburg Burnout Inventory in Nursing, S-BAT = Student version of the Burnout Assessment Tool, SBI = School-Burnout Inventory, SBI-U = School Burnout Inventory-Universitarios, Short version of BAT-C = Short version of the emotional burnout assessment instrument (BAT-C), SSBS = Secondary School Burnout Scale, ULB = Undergraduate Learning Burnout scale, UW-ESBS = University of Windsor Employed Student Burnout Survey